

Data Collection and Preprocessing Phase

Date	3-7-2026
Roll ID	TYA105
Project Title	Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report
Maximum Marks	2 Marks

Data Exploration and Preprocessing Template

Section	Description
Data Overview	The dataset contains 1,360 records and 16 attributes representing AI adoption in education across multiple countries and regions from 2015 to 2026.
Data Cleaning	Checked for missing values, duplicate records, invalid entries, and inconsistencies. Missing values in the Government AI Policy field were handled appropriately.
Data Transformation	Filtered unnecessary records, standardized categorical values, created calculated fields, and prepared the dataset for Tableau dashboards.
Data Type Conversion	Verified and corrected numeric, percentage, and categorical data types before visualization.
Column Splitting & Merging	Examined the dataset for columns requiring transformation. Existing columns were used without major splitting or merging.
Handling Missing Values	Missing values were identified and managed by retaining null values or categorizing them as "Unknown" where required.
Duplicate Removal	Verified that the dataset contained no duplicate records.
Data Validation	Cross-checked row count, column names, percentage values, and calculated fields before dashboard development.
Exploratory Analysis (EDA)	Data Analyzed country-wise AI adoption, regional trends, AI tool usage, urban-rural differences, student-teacher usage, and yearly growth patterns.

Section	Description
Final Output	The cleaned dataset was successfully imported into Tableau and used to develop interactive dashboards and stories.

❖ **Summary of Preprocessing-**

- **Dataset:** Global AI in Education
- **Rows:** 1,360
- **Columns:** 16
- **Missing Values:** 441 (Government AI Policy)
- **Duplicate Records:** 0
- **Software Used:** Tableau Public, Microsoft Excel
- **Final Status:** Dataset successfully cleaned and prepared for visualization.